

Algal-oil capsules and cooked salmon: nutritionally equivalent sources of docosahexaenoic acid.

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Food and nutrition professionals question whether supplement-sourced nutrients appear to be equivalent to those derived from natural food sources. We compared the nutritional availability of docosahexaenoic acid (DHA) from algal-oil capsules to that from assayed cooked salmon in 32 healthy men and women, ages 20 to 65 years, in a randomized, open-label, parallel-group study. In this 2-week study comparing 600 mg DHA/day from algal-oil capsules to that from assayed portions of cooked salmon, mean change from baseline in plasma phospholipids and erythrocyte DHA levels was analyzed and DHA levels were compared by Student's t tests. In post-hoc analyses to determine bioequivalence, least-squares mean ratios of percent change from baseline in plasma phospholipid and erythrocyte DHA levels were compared. DHA levels increased by approximately 80% in plasma phospholipids and by approximately 25% in erythrocytes in both groups. Changes in DHA levels in plasma phospholipids and erythrocytes were similar between groups. As measured by delivery of DHA to both plasma and erythrocytes, fish and algal-oil capsules were equivalent. Both regimens were generally well-tolerated. These results indicate that algal-oil DHA capsules and cooked salmon appear to be bioequivalent in providing DHA to plasma and red blood cells and, accordingly, that algal-oil DHA capsules represent a safe and convenient source of non-fish-derived DHA.